BTS–III–11.18–1727

Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

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B.Tech. Degree III Semester Examination November 2018

CS 15-1305 PRINCIPLES OF PROGRAMMING LANGUAGES

(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) What do you mean by unambiguous grammar? Give an unambiguous grammar for arithmetic expression.
(b) Define attribute grammar.
(c) Give an example for the orthogonality characteristic of programming language.
(d) Define life time of a variable. Illustrate with example.
(e) What is a coroutine?
(f) What is polymorphism? Give an example of polymorphism in C++.
(g) What is abstract data type?
(h) How information hiding is achieved in C++?
(i) What are Horn Clauses?
(j) What is lambda calculus?

PART B

$(4 \times 10 = 40)$

II. Explain the criteria for language evaluation. (10)

OR

III. (a) Describe denotational semantics. (5)
(b) Give the denotational semantics for a binary number by specifying the grammar. (5)

IV. What are the various parameter passing mechanisms in the programming language design with suitable example? (10)

OR

V. (a) What are named constants? Give example. (5)
(b) Explain how a variable can be characterized. (5)

VI. Explain the exception handling mechanism in C++. Give example. (10)

OR

VII. What are the various types of inheritance in C++? Illustrate with examples. (10)

VIII. (a) Write a function for the factorial in LISP. (5)
(b) Explain any 5 functions in LISP. (5)

OR

IX. (a) What are the basic elements in PROLOG? (5)
(b) Explain the tracing model in PROLOG. (5)

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